

AI-Assistant

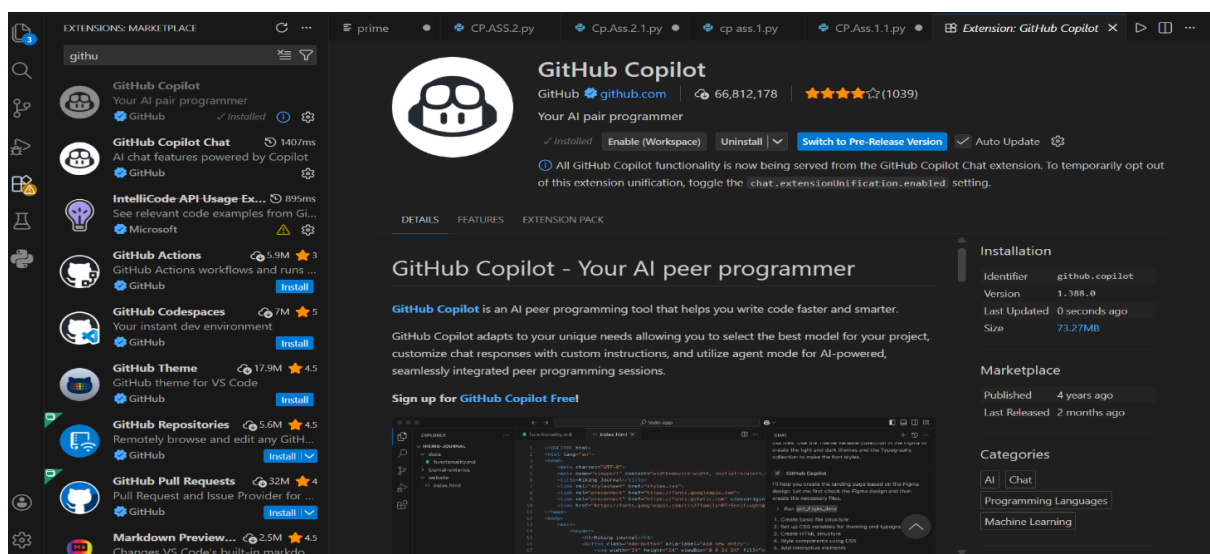
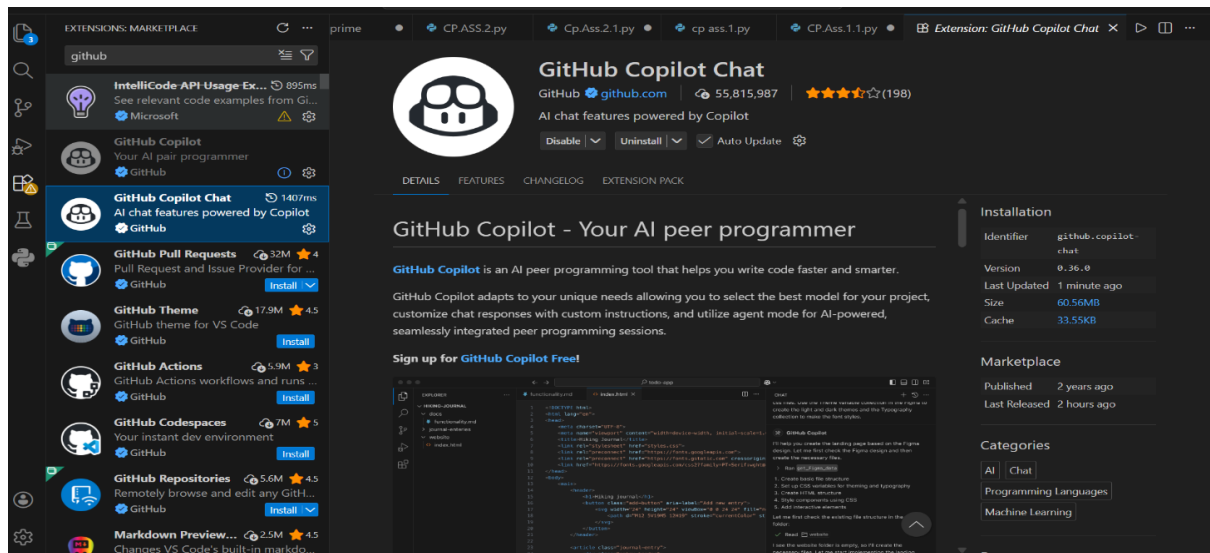
Batch -14

2303A510A0(VISHNU VARDHAN)

Lab Assisgnment-1.4

Task -0: Install and configure GitHub Copilot in VS Code. Take screenshots of each step

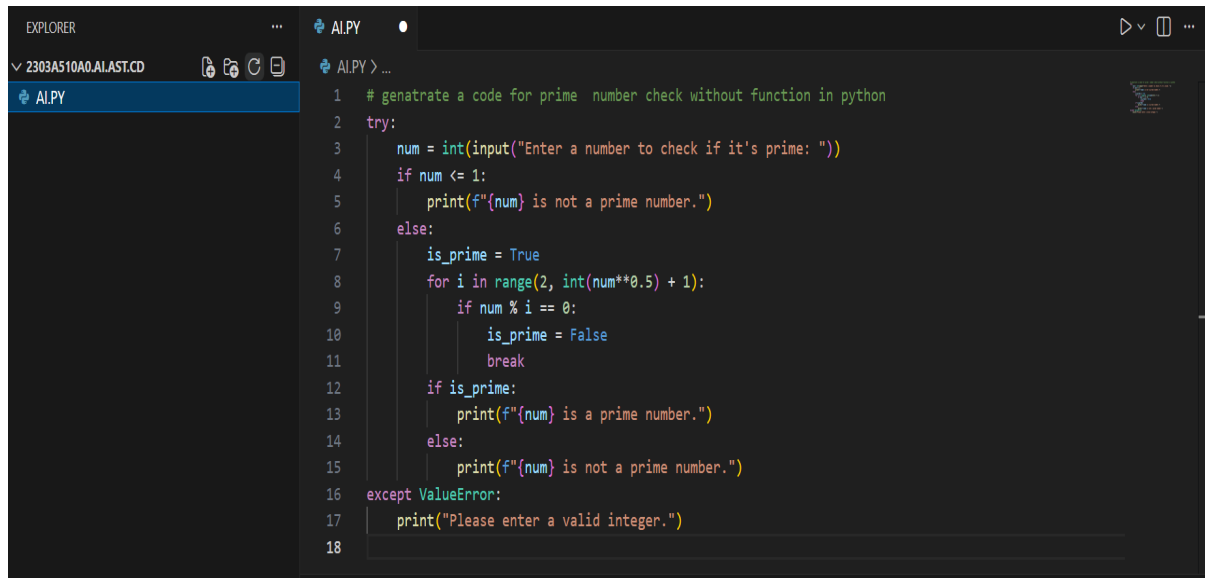
Output:



Task 1: AI-Generated Logic Without Modularization (Prime Number Check Without Functions)

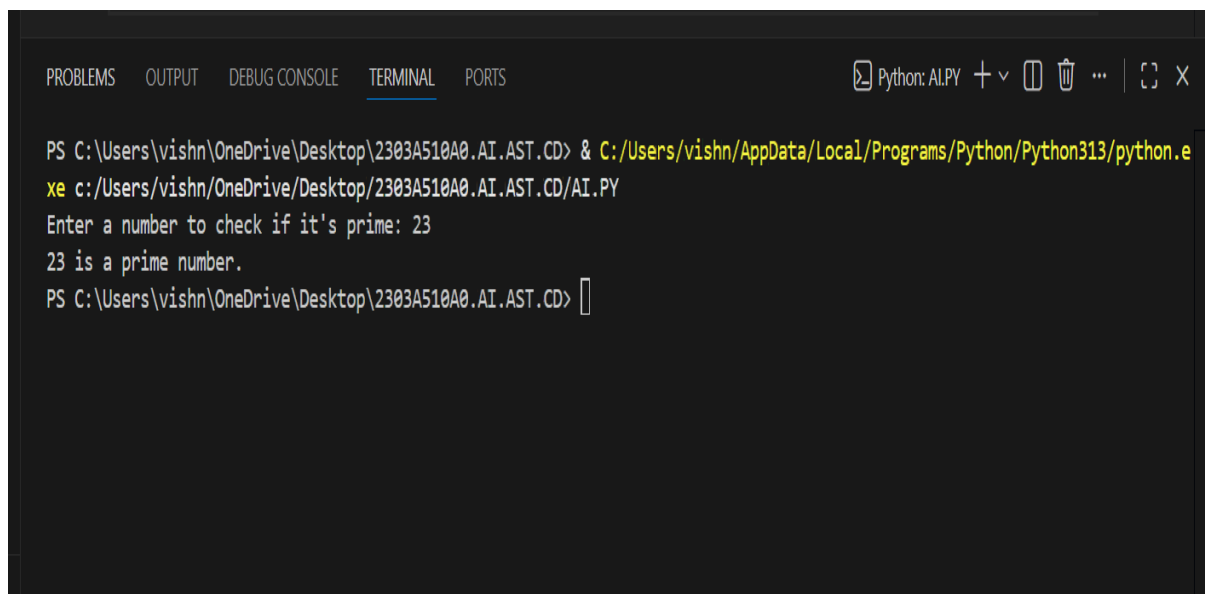
#Generate a code for prime number check without function in python

Code:



```
1 # genatrare a code for prime number check without function in python
2 try:
3     num = int(input("Enter a number to check if it's prime: "))
4     if num <= 1:
5         print(f"{num} is not a prime number.")
6     else:
7         is_prime = True
8         for i in range(2, int(num**0.5) + 1):
9             if num % i == 0:
10                 is_prime = False
11                 break
12         if is_prime:
13             print(f"{num} is a prime number.")
14         else:
15             print(f"{num} is not a prime number.")
16 except ValueError:
17     print("Please enter a valid integer.")
18
```

Output:

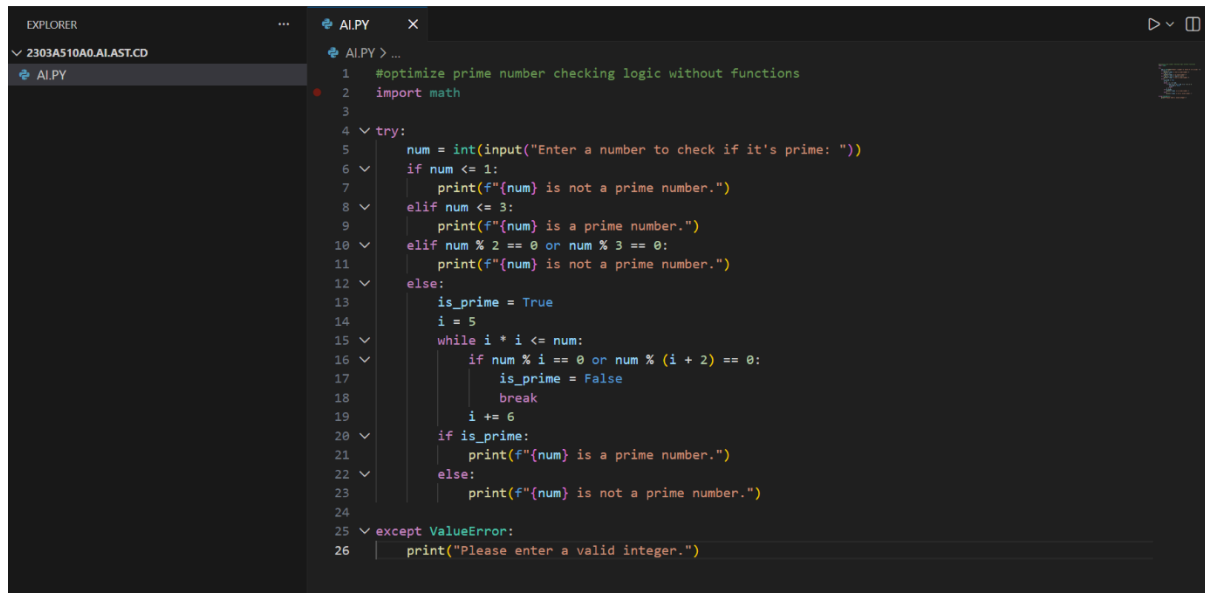


```
PS C:\Users\vishn\OneDrive\Desktop\2303A510A0.AI.AST.CD> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe c:/Users/vishn/OneDrive/Desktop/2303A510A0.AI.AST.CD/AI.PY
Enter a number to check if it's prime: 23
23 is a prime number.
PS C:\Users\vishn\OneDrive\Desktop\2303A510A0.AI.AST.CD>
```

Task 2: Efficiency & Logic Optimization (Cleanup)

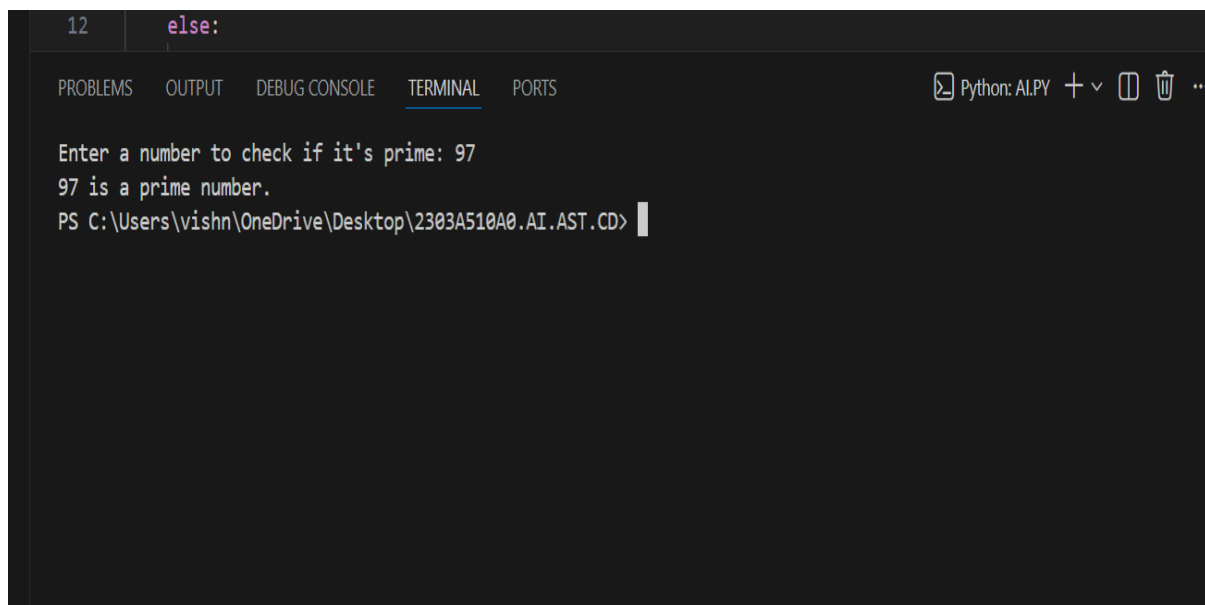
#optimize prime number checking logic without functions

Code:



```
1 #optimize prime number checking logic without functions
2 import math
3
4 try:
5     num = int(input("Enter a number to check if it's prime: "))
6     if num <= 1:
7         print(f"{num} is not a prime number.")
8     elif num <= 3:
9         print(f"{num} is a prime number.")
10    elif num % 2 == 0 or num % 3 == 0:
11        print(f"{num} is not a prime number.")
12    else:
13        is_prime = True
14        i = 5
15        while i * i <= num:
16            if num % i == 0 or num % (i + 2) == 0:
17                is_prime = False
18                break
19            i += 6
20        if is_prime:
21            print(f"{num} is a prime number.")
22        else:
23            print(f"{num} is not a prime number.")
24
25 except ValueError:
26     print("Please enter a valid integer.")
```

Output:

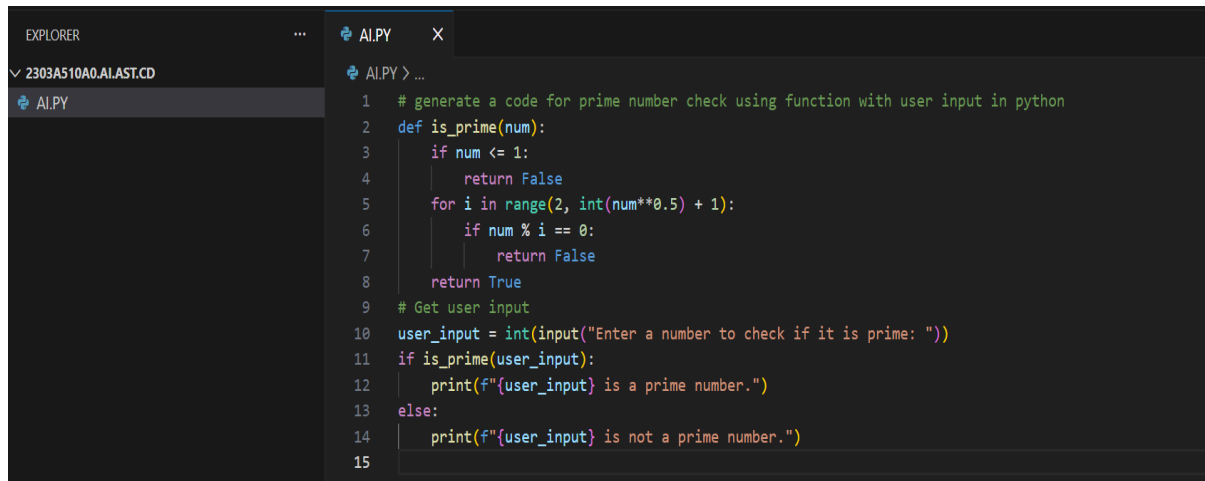


```
12 else:
Enter a number to check if it's prime: 97
97 is a prime number.
PS C:\Users\vishn\OneDrive\Desktop\2303A510A0.AI.AST.CD>
```

Task 3: Modular Design Using AI Assistance (Prime Number Check Using Functions)

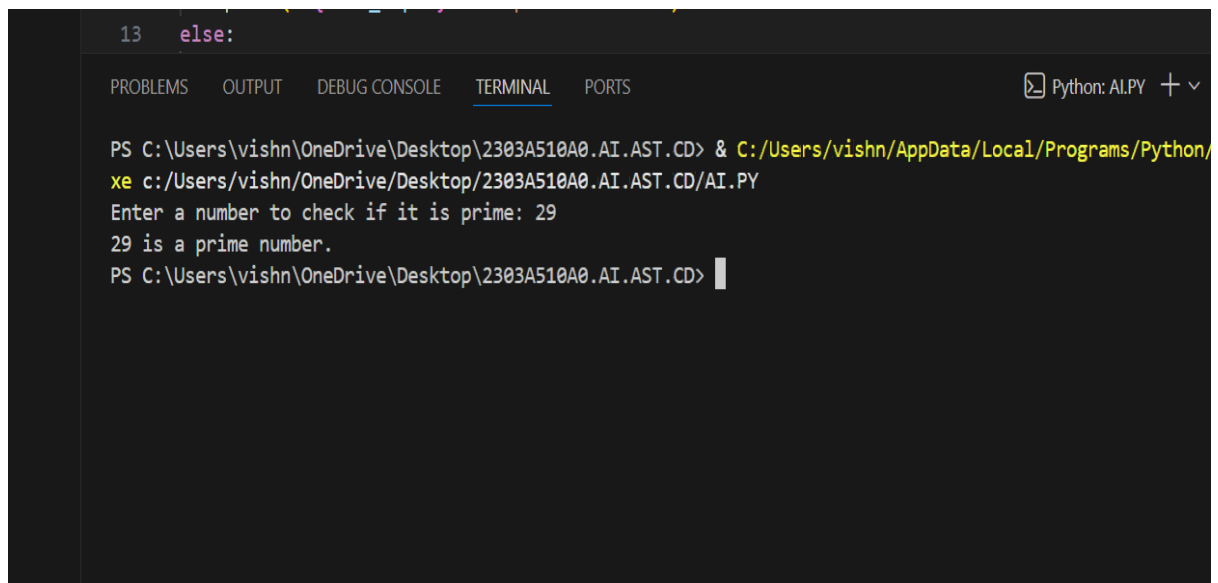
Generate a code for prime number check using function with user input in python

Code:



```
1 # generate a code for prime number check using function with user input in python
2 def is_prime(num):
3     if num <= 1:
4         return False
5     for i in range(2, int(num**0.5) + 1):
6         if num % i == 0:
7             return False
8     return True
9 # Get user input
10 user_input = int(input("Enter a number to check if it is prime: "))
11 if is_prime(user_input):
12     print(f"{user_input} is a prime number.")
13 else:
14     print(f"{user_input} is not a prime number.")
15
```

Output:



```
13 else:
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python: AI.PY + v
PS C:\Users\vishn\OneDrive\Desktop\2303A510A0.AI.AST.CD> & C:/Users/vishn/AppData/Local/Programs/Python/
xe c:/Users/vishn/OneDrive/Desktop/2303A510A0.AI.AST.CD/AI.PY
Enter a number to check if it is prime: 29
29 is a prime number.
PS C:\Users\vishn\OneDrive\Desktop\2303A510A0.AI.AST.CD>
```